Lecture 11 — Exotic Options: Overview and Numerical Procedures

Computational Finance (WI4151), TU Delft — exam notes

Scope. This is the closing, survey lecture. It is deliberately broad rather than deep on any single product, but the slides do carry real technical content: the Fouque–Han stochastic-volatility Asian PDE and the Vecer dimension reduction are worked out in detail, and several closed forms (geometric Asian, digital $= e^{-rT}N(d_2)$) appear or are implied. The aim of these notes is to fix, for each exotic, three things that an examiner will probe: *(i)* the precise payoff, *(ii)* the economic risk/need it serves, and *(iii)* which numerical method is appropriate and *why*. Barrier options are listed in the lecture but explicitly deferred to Lecture 10; see `notes/Lecture10-notes.pdf`.

1 Why exotic options exist

Vanilla European calls/puts pay off a function of the terminal price S_T alone. Many real exposures are not of that form, and the lecture organises the rationale for exotics under five heads.

(1) **Customised payoff structure.** A European exporter wants protection only in a specific tail: pay a fixed amount if EUR/USD > 1.10 at maturity, nothing otherwise. This discontinuous (digital) payoff cannot be replicated cheaply with vanilla calls — a tight call spread approximating it requires an unbounded notional as the spread width $\to 0$ (see §4).

(2) **Reduction of hedging cost.** An airline hedging one month of jet-fuel procurement buys a *single* Asian call on the monthly average rather than ~ 20 daily vanilla calls. Averaging lowers the effective volatility of the underlying quantity, hence lowers the premium materially.

(3) **Better matching of business exposure.** The airline's economic exposure is the average procurement cost $\frac{1}{n}\sum_{i=1}^{n} S_{t_i}$, not the terminal price S_T. An Asian option matches this exactly; a terminal-price call would over- or under-hedge for most of the month.

(4) **Yield enhancement in low-rate environments.** A worst-of autocallable on {EURO STOXX 50, S&P 5 pays a 10% annual coupon if the worst performer is above a barrier, with possible early redemption. The investor implicitly *sells* correlation and downside risk on the weakest index in return for an above-market coupon.

(5) **Volatility / correlation trading.** A worst-of put gives clean exposure to the correlation risk premium (the issuing bank is short correlation); a cliquet gives forward-skew / forward-volatility exposure. These are first-order exposures a vanilla portfolio cannot replicate cleanly.

Who uses them. Corporates (FX/commodity/rate hedging), asset managers, insurers (cliquet-style equity-linked guarantees), private banks (yield-enhancement notes funded by implicitly selling option-like risks), structured-product desks (autocallables, reverse convertibles), and commodity/FX hedgers (Asian, barrier, digital).

Pre-2008 vs. post-2008

Before 2008: cheap funding, weak regulatory capital constraints, large OTC growth, aggressive yield-enhancement demand, heavy structured-product issuance. Popular: barriers, reverse convertibles, callable range accruals, snowballs, autocallables, long-dated path-dependent and highly bespoke OTC exotics. Theme: banks optimised yield and leverage aggressively.
Lessons of 2008: counterparty risk became critical; liquidity assumptions failed; *correlation structures broke down* (directly hitting basket/worst-of books); model risk became visible; funding costs mattered.
After 2008: XVA adjustments (CVA, FVA, MVA), Basel III capital constraints, increased collateralisation, simplification of product books, and a preference for hedgeable flow products. The surviving-product table is worth memorising:

Category	Before 2008	After 2008
Highly bespoke OTC	very common	reduced
Barriers	very active	still active
Asians	active	active
Digitals	active	active
Basket / worst-of	growing	*very active*
Cliquets	active (aggressive)	reduced (insurance-driven)
Long-dated exotics	common	reduced
Autocallables	growing	*dominant in equity SPs*

The driving trend: markets now prefer products that are easier to hedge, easier to model, easier to explain to regulators, and more liquid. The two products that *grew* (basket/worst-of, autocallables) did so because retail yield demand outran the correlation-modelling caveats.

2 Asian options

Payoff. Asian payoffs depend on the average of the underlying over monitoring dates $t_1, \dots, t_n$. *Fixed-strike arithmetic-average call:*

$$\Big(\tfrac{1}{n}\textstyle\sum_{i=1}^{n} S_{t_i} - K\Big)^+, \qquad \textit{geometric-average call:} \quad \Big(\big(\textstyle\prod_{i=1}^{n} S_{t_i}\big)^{1/n} - K\Big)^+.$$

Floating-strike variants replace K by a multiple of S_T. Asians are path-dependent, carry *reduced volatility exposure* (the average is less volatile than the terminal value), are *cheaper* than the corresponding vanilla, and are popular in commodities and FX where the operational exposure is to an average and where averaging reduces fixing-manipulation risk.

Why the geometric average has a closed form (worked). Under GBM, $S_{t_i} = S_0 \exp\big((r - q - \frac{1}{2}\sigma^2)t_i + \sigma W_{t_i}\big)$. The geometric average is

$$G = \Big(\textstyle\prod_{i=1}^{n} S_{t_i}\Big)^{1/n} = S_0 \exp\Big(\tfrac{1}{n}\textstyle\sum_{i=1}^{n}\big[(r - q - \tfrac{1}{2}\sigma^2)t_i + \sigma W_{t_i}\big]\Big),$$

i.e. $\ln G$ is an affine function of the jointly Gaussian vector $(W_{t_1}, \dots, W_{t_n})$, hence $\ln G$ is itself normal. Therefore G is *lognormal* and the geometric Asian prices by a Black–Scholes-type formula. With $\bar{t} = \frac{1}{n}\sum_i t_i$ and using $\mathrm{Cov}(W_{t_i}, W_{t_j}) = \min(t_i, t_j)$,

$$\mu_G \equiv \mathbb{E}[\ln G] = \ln S_0 + (r - q - \tfrac{1}{2}\sigma^2)\bar{t}, \qquad \sigma_G^2 \equiv \mathrm{Var}(\ln G) = \frac{\sigma^2}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n}\min(t_i, t_j).$$

For equally spaced dates $t_i = i\,\Delta t$, $\Delta t = T/n$, one gets the standard

$$\bar{t} = \frac{(n+1)}{2n}T, \qquad \sigma_G^2 = \sigma^2\,\Delta t\,\frac{(n+1)(2n+1)}{6n^2} \xrightarrow[n\to\infty]{} \frac{\sigma^2 T}{3}.$$

Define an effective volatility $\hat\sigma^2 = \sigma_G^2/T$ and an effective dividend $\hat q$ such that $\mathbb{E}[G] = S_0 e^{(r-\hat q)T}$, i.e. $\hat q = r - \frac{1}{T}(\mu_G - \ln S_0) - \frac{1}{2}\hat\sigma^2$. Then the geometric Asian call is the Black–Scholes call price $e^{-rT}\big[\mathbb{E}[G]\,N(d_1) - K\,N(d_2)\big]$ with

$$d_{1,2} = \frac{\ln(\mathbb{E}[G]/K) \pm \frac{1}{2}\hat\sigma^2 T}{\hat\sigma\sqrt{T}}.$$

Key qualitative facts: (a) the continuous-monitoring variance is $\sigma^2 T/3$, one-third of the vanilla variance — this is exactly why the Asian premium is lower; (b) the geometric Asian is the canonical *control variate* for Monte Carlo pricing of the arithmetic Asian (Kemna–Vorst 1990).

Why arithmetic Asians have no closed form. A sum of lognormals is not lognormal, so $A_T = \frac{1}{n}\sum_i S_{t_i}$ has no elementary density and its characteristic function (CF) is *not closed-form even under affine models.* This is the central numerical difficulty and dictates the method choice below.

Numerical methods and why.

- *Moment matching (Turnbull–Wakeman, Levy).* Approximate A_T by a lognormal with the same first two moments. Compute $m_1 = \mathbb{E}[A_T] = \frac{1}{n}\sum_i F(0,t_i)$ (forwards) and $m_2 = \mathbb{E}[A_T^2] = \frac{1}{n^2}\sum_i\sum_j \mathbb{E}[S_{t_i}S_{t_j}]$, fit (μ_A, σ_A), and price with a Black-style formula. Fast and accurate for moderate volatility, but it is an *approximation* and degrades in the tails / at long maturity where higher cumulants matter.

- *PDE.* The running average A_t is an extra state, giving a 2-D PDE in (S, A) with terminal $V(T,S,A) = (A-K)^+$. Vecer's similarity/numeraire trick (1995/2001) reduces this *exactly* to a 1-D PDE; see §3.

- *COS / Fourier (Zhang–Oosterlee 2013).* Either (i) approximate ϕ_{A_T} via cumulants and apply the European COS quadrature, or (ii) run *backward COS recursion* over the monitoring dates using the one-step conditional CF. The COS price has the familiar form

$$V_0 = e^{-rT}\sum_{k=0}^{N-1\,'} \mathrm{Re}\Big[\phi_{A_T}\Big(\tfrac{k\pi}{b-a}\Big)e^{-ik\pi a/(b-a)}\Big]H_k,$$

  with payoff cosine coefficients H_k and the truncation $[a,b]$ chosen from the *cumulants of A_T*. The same $[a,b]$ pitfall as for vanilla COS applies: too narrow an interval at short maturity (small c_2) loses mass.

- *Monte Carlo.* Most flexible, natural for path dependence: simulate M GBM paths, form $A_T^{(m)}$, discount $X^{(m)} = e^{-rT}(A_T^{(m)} - K)^+$, average, and report a CI from $s_X/\sqrt{M}$. *Variance reduction* (geometric-Asian control variate) is essentially mandatory.

3 Asian under stochastic volatility: Fouque–Han PDE and Vecer reduction

The slides develop the constant-vol PDE all the way to a stochastic-volatility model, so the examiner may push here. The reference is Fouque & Han (2003).

Model. Risk-neutral dynamics with a mean-reverting volatility factor Y:

$$\mathrm{d}S_t = rS_t\,\mathrm{d}t + f(Y_t)S_t\,\mathrm{d}W_t^*, \qquad \mathrm{d}Y_t = \big[\alpha(m - Y_t) - \beta\Lambda(Y_t)\big]\,\mathrm{d}t + \beta\big(\rho\,\mathrm{d}W_t^* + \sqrt{1-\rho^2}\,\mathrm{d}Z_t^*\big),$$

with W^*, Z^* independent, $\sigma_t = f(Y_t)$, leverage ρ, and Λ the combined market price of risk. Path dependence enters through the running integral $I_t = \int_0^t S_u\,\mathrm{d}u$, $\mathrm{d}I_t = S_t\,\mathrm{d}t$. The triple (S_t, Y_t, I_t) is Markov.

3-D pricing PDE. For the floating-strike Asian call $P = \mathbb{E}^*[e^{-r(T-t)}(S_T - I_T/T)^+ \mid \cdot]$, Feynman–Kac gives

$$\partial_t P + \tfrac{1}{2} f^2 s^2 \partial_{ss} P + \rho\beta f s\, \partial_{sy} P + \tfrac{1}{2}\beta^2 \partial_{yy} P + r(s\,\partial_s P - P) + \big(\alpha(m-y) - \beta\Lambda\big)\partial_y P + s\,\partial_I P = 0,$$

terminal $P(T, s, y, I) = (s - I/T)^+$. Read the four blocks: second-order (s, y) diffusion (cross term carries ρ); first-order risk-neutral drift in s and risk-adjusted drift in y; the pure-drift term $s\,\partial_I P$ (since $\mathrm{d}I = S\,\mathrm{d}t$); and $-rP$ discounting. *Issue:* three spatial dimensions $\Rightarrow$ expensive grids.

Vecer reduction (why it works). Replicate $\frac{1}{T}\int_0^T S_u\,\mathrm{d}u$ by a *self-financing* portfolio holding a *time-deterministic* number of shares $q(t) = \frac{1-e^{-r(T-t)}}{rT}$, with wealth $\mathrm{d}X_t = rX_t\,\mathrm{d}t + q(t)(\,\mathrm{d}S_t - rS_t\,\mathrm{d}t)$. Choosing $X_0 = q(0)S_0 - e^{-rT}K_2$ integrates to $X_T = \frac{1}{T}\int_0^T S_u\,\mathrm{d}u - K_2$, so the unified payoff $h\big(\frac{1}{T}\int_0^T S_u\,\mathrm{d}u - K_1 S_T - K_2\big) = h(X_T - K_1 S_T)$ covers fixed-strike ($K_1 = 0$) and floating-strike ($K_2 = 0$) Asians. The point: trading the deterministic $q(t)$ *collapses the path-dependent state I_t into the single stochastic state X_t.* Switching to the stock numeraire ($\Psi_t = X_t/S_t$) and using the positive homogeneity $h(xy) = xh(y)$ of calls/puts reduces the 3-D PDE to a 2-D PDE in (ψ, y); in the constant-vol limit Y drops and one recovers the celebrated *1-D Vecer Asian PDE.* (Caveat: this prices the fresh Asian at $t = 0$; a seasoned Asian uses Asian put–call parity.)

4 Digital / binary options

Payoff. *Cash-or-nothing call:* $\mathbf{1}_{\{S_T>K\}}$ (times a notional Q). *Asset-or-nothing call:* $S_T\mathbf{1}_{\{S_T>K\}}$. The payoff is discontinuous at K. Digitals are a building block of structured products (range accruals, autocallables, structured deposits) and are directly tied to the risk-neutral CDF.

Closed form (derive). A cash-or-nothing call pays \$1 on $\{S_T > K\}$, so

$$C_{\text{dig}} = e^{-rT}\,\mathbb{Q}(S_T > K).$$

Under GBM, $\ln S_T = \ln S_0 + (r - q - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}\,Z$, $Z \sim N(0,1)$, hence $\{S_T > K\} = \{Z > -d_2\}$ with

$$d_2 = \frac{\ln(S_0/K) + (r - q - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \qquad \mathbb{Q}(S_T > K) = \mathbb{Q}(Z > -d_2) = N(d_2).$$

Therefore $C_{\text{dig}} = e^{-rT}N(d_2)$ (notional Q: multiply by Q). A digital *prices a risk-neutral probability.* The asset-or-nothing call equals $S_0 e^{-qT}N(d_1)$, and the vanilla call decomposes as (asset-or-nothing) $-K\times$(cash-or-nothing), recovering $C = S_0 e^{-qT}N(d_1) - Ke^{-rT}N(d_2)$.

Static-replication / Breeden–Litzenberger link. Differentiating the vanilla call price in strike,

$$\frac{\partial C}{\partial K} = -e^{-rT}\mathbb{Q}(S_T > K) = -C_{\text{dig}},$$

so a digital is (minus) the strike-derivative of vanilla prices — this connects digitals to the implied distribution, local-vol extraction, and static replication. Concretely, approximate

$$\mathbf{1}_{\{S_T>K\}} \approx \frac{(S_T - K + \frac{\varepsilon}{2})^+ - (S_T - K - \frac{\varepsilon}{2})^+}{\varepsilon}:$$

buy $1/\varepsilon$ calls at $K - \frac{\varepsilon}{2}$, sell $1/\varepsilon$ calls at $K + \frac{\varepsilon}{2}$. *Centring* around K gives second-order accuracy in ε; a one-sided spread gives only first-order.

The hedging-blow-up (this is the examiner's favourite). The closed form is clean, but the *hedge* is not. The notional of the replicating call spread is $1/\varepsilon \to \infty$ as the spread tightens. Equivalently, near expiry and near the strike the digital's delta spikes (Dirac-like) and gamma flips sign — "*pin risk*". A bank delta-hedging a sold digital must trade a huge, sign-changing position in a vanishingly small neighbourhood of K as $t \to T$, which is impossible to execute cleanly; in practice desks over-hedge by pricing/hedging a *call spread* (a slightly conservative super-replicating digital) rather than the exact digital. This is why the "closed form exists" fact is misleading: the difficulty is risk management, not valuation.

Numerical methods and why.

- *Closed form / Black–Scholes:* exact under GBM as above.
- *COS / Fourier (Fang–Oosterlee 2008): very* well suited — the indicator payoff has closed-form cosine coefficients, and a digital is just the original European COS method with a different H_k. No discontinuity penalty in the payoff side, since the cosine coefficients are computed analytically.
- *PDE:* terminal condition $V(T,S) = \mathbf{1}_{\{S>K\}}$ has a jump; one must align the grid to K or smooth the discontinuity, then step back with an implicit / Crank–Nicolson scheme (Crank–Nicolson can oscillate near the kink, so smoothing/Rannacher start-up helps).
- *Monte Carlo:* trivial to code ($e^{-rT}\mathbf{1}_{\{S_T>K\}}$) but the indicator makes the estimator high-variance near K and the pathwise Greeks ill-defined; use conditional MC / smoothing.

5 Basket and worst-of options

Payoff. *Basket call:* $\left(\sum_{i=1}^{n} w_i S_i(T) - K\right)^+$ on a weighted sum. *Worst-of call:* $\left(\min_i S_i(T) - K\right)^+$ on the worst performer. Market term sheets express everything in *performances* $S_i(T)/S_i(0)$ so strikes are percentages and assets at different absolute levels are comparable. The dominant risk factor is *correlation.*

Remark 1 (textbook vs. traded structure)**.** *The clean "call on the min" is rarely the traded payoff. In a worst-of autocallable the embedded option is almost always a worst-of* put*: the investor is* short *downside on the weakest index (e.g. redemption reduced if* $\min_i S_i(T)/S_i(0)$ *falls below a* 60% *barrier). The high coupon is funded by the investor selling that correlation/downside risk. Be ready to state this distinction.*

Why correlation dominates (worked for the basket). For the basket $B_T = \sum_i w_i S_i(T)$,

$$\mathrm{Var}(B_T) = \sum_i \sum_j w_i w_j \,\mathrm{Cov}(S_i, S_j) = \sum_i w_i^2 \,\mathrm{Var}(S_i) + \sum_{i \neq j} w_i w_j \rho_{ij} \sqrt{\mathrm{Var}(S_i)\mathrm{Var}(S_j)}.$$

The off-diagonal ρ_{ij} terms drive the variance: a basket of *positively* correlated assets is more volatile (so a call on it is dearer) than one of weakly/negatively correlated assets, which diversify. A worst-of is even more correlation-sensitive: as correlation $\to 1$ the worst-of approaches a single-asset option (less downside), and as correlation falls the minimum is dragged down by idiosyncratic moves (more downside, so a worst-of put is dearer). The 2008 break-down of correlation structure is precisely what made these books dangerous.

Numerical methods and why. The challenge is *multi-dimensionality.*

- *Moment matching / lognormal approximation:* approximate B_T by a single lognormal with $m_1 = \sum_i w_i F_i(0,T)$ and $m_2 = \sum_i \sum_j w_i w_j \mathbb{E}[S_i(T)S_j(T)]$, then a Black-style formula. Cheap, but only for the (positively weighted) basket; it cannot handle the min in a worst-of and breaks for low/negative weights.

- *PDE:* dimension-n grid with correlation cross-terms $\rho_{ij}\sigma_i\sigma_j S_i S_j\, \partial_{S_i S_j} V$. *Curse of dimensionality*: cost grows like M^n, so practical only for $n \leq 2$–3.

- *COS / Fourier:* efficient under an affine *joint* CF. Concretely, 2-D COS for a 2-asset basket / call-on-min (Ruijter–Oosterlee 2012), and a 3-asset spread (Pellegrino–Sabino 2016). For ≥ 4 assets *no dedicated COS paper exists* — the tensor-product cosine grid suffers the same curse of dimensionality as the PDE. [**Honest gap: the slides explicitly flag that no COS method is available for high-dimensional worst-of/basket.**]

- *Monte Carlo:* the *industry standard.* Estimate the volatility vector and correlation matrix, Cholesky-factor $\Sigma = LL^\top$, generate independent normals Z, correlate by $\tilde{Z} = LZ$, simulate each asset, and evaluate e.g. $X^{(m)} = e^{-rT}(\min_i S_i^{(m)}(T) - K)^+$. MC handles arbitrary dimension and correlation naturally; enhance with antithetics, quasi-MC, control variates. This is *the* reason MC dominates the multi-asset book.

6 Cliquet options

Payoff. A cliquet ("ratchet") locks in periodic, capped/floored returns. With period returns $R_i = S_{t_i}/S_{t_{i-1}} - 1$, local floor F and cap C,

$$\text{Payoff} = \sum_{i=1}^{n} \min\big(\max(R_i, F),\, C\big), \qquad \ell_i = \min(\max(R_i, F), C).$$

A global cap/floor may be applied to the accumulated sum $G = \sum_i \ell_i$. Cliquets are path-dependent (via the resets), carry a *periodic reset / ratchet* feature, are highly sensitive to *forward skew / forward volatility* and to *local-volatility* dynamics, and are a workhorse of insurance and capital-protected notes (e.g. principal-guaranteed with an 8% annual cap and 0% floor — a smooth, behaviourally appealing return profile).

Why the structure matters. A cliquet is essentially a strip of *forward-starting* capped/floored options: each ℓ_i is determined over the future window $[t_{i-1}, t_i]$, so its value depends on the *forward* volatility and forward skew prevailing then, not on today's spot-starting smile. This is precisely the exposure a vanilla portfolio (whose options all start today) cannot replicate, and it is why cliquet pricing is so sensitive to the smile-dynamics assumption: a model that fits today's smile but extrapolates forward skew poorly will misprice the cliquet. The reset feature also makes the product *vega-rich across reset dates.*

Numerical methods and why.

- *Monte Carlo:* simulate at the reset dates $t_0, \ldots, t_n$, form $R_i^{(m)}$, apply $\ell_i^{(m)} = \min(\max(R_i^{(m)}, F), C)$, aggregate $G^{(m)} = \sum_i \ell_i^{(m)}$, apply any global cap/floor, then $\hat{V}_0 = \frac{1}{M}\sum_m e^{-rT} G^{(m)}$. *Widely used*, because it combines trivially with local-vol, stochastic-vol and hybrid models — the very models needed to get the forward skew right.

- *PDE:* reset features create *jump conditions*; carry the accumulated-return state G, solve between reset dates, and at each t_i update $G_i = G_{i-1} + \ell_i$. A global cap/floor adds a state variable.

- *COS / Fourier:* efficient for the periodic structure — one CF per reset period (forward log-return), expand the capped/floored return payoff in a cosine series, and recursively aggregate the resets. [**Honest gap: the slides note no dedicated COS paper for the pure cliquet derivative; the closest is COS for cliquet-style equity-indexed annuities (Deng, Dulaney, McCann & Yan, 2017); an alternative Fourier route is PROJ / frame-duality (Kirkby–Nguyen), which is *not* the COS method.**]

- *Tree / lattice:* pedagogically useful, impractical in higher dimensions.

7 Barrier options (cross-reference)

Barriers are listed as a surviving, still-active exotic but the lecture explicitly defers them to Lecture 10. *In one line for completeness:* a knock-in/knock-out call pays the vanilla payoff only if the path does/does not breach a barrier H before maturity. The appropriate method is the COS backward recursion over monitoring dates (the cosine coefficients are restricted to the survival region $[\cdot]$ at each step), with Richardson extrapolation in the number of monitoring dates to recover the continuous-barrier limit; see `notes/Lecture10-notes.pdf` for the full treatment of the truncation range, the χ/ψ integrals on the survival interval, and the discrete-to-continuous correction.

8 Numerical-method map (the takeaway table)

Exotic	First-choice method	Why
Geometric Asian	closed form	sum of normals $\Rightarrow$ lognormal G
Arithmetic Asian	MC (+ geo. control variate) / 1-D Vecer PDE	no closed-form CF; path-depend
Digital	closed form $e^{-rT}N(d_2)$ / COS	probability under $\mathbb{Q}$; indicator ha
Basket ($n \le$ 2–3)	2-D COS / moment matching	affine joint CF available in low d
Basket / worst-of ($n \ge 4$)	Monte Carlo	curse of dimensionality kills PDE
Cliquet	MC (local/stoch-vol)	forward-skew sensitive; reset jum
Barrier	COS backward recursion (Lec. 10)	monitoring-date survival integral

Course-level conclusions. (i) Exotics exist to match real exposures vanillas cannot. (ii) Post-2008 markets shifted toward simpler, hedgeable, modellable, liquid products. (iii) Asians, digitals, baskets/worst-of and cliquets remain relevant. (iv) Numerical methods are central because closed forms are usually unavailable — and *Monte Carlo remains dominant in industry* precisely because of path dependence, high dimensionality, and the need for local/stochastic-vol models. (v) Modern themes: correlation risk, volatility-smile dynamics, XVA/funding, regulatory capital.

9 Likely examiner questions (with the trap)

- *"Why is the geometric Asian closed-form but the arithmetic one not?"* Sum of lognormals is not lognormal; product is. Give $\sigma_G^2 \to \sigma^2 T/3$.

- *"A digital has a closed form $e^{-rT}N(d_2)$ — so why is it considered hard?"* The *valuation* is easy; the *hedge* blows up (delta spike, gamma sign-flip, pin risk near K at expiry; replicating call-spread notional $1/\varepsilon \to \infty$). Desks hedge a call spread, not the exact digital.

- *"Why Monte Carlo for a 5-asset worst-of and not COS?"* Curse of dimensionality: tensor cosine grid / PDE grid cost $\sim M^n$. No COS paper beyond 3 assets.

- *"What is the single most important risk in a basket/worst-of, and show it."* Correlation; $\mathrm{Var}(B_T) = \sum_i \sum_j w_i w_j \mathrm{Cov}(S_i, S_j)$, off-diagonal terms dominate.

- *"Why is COS natural for the arithmetic Asian despite no closed-form CF?"* Approximate ϕ_{A_T} via cumulants (set $[a, b]$ from cumulants) or recurse over monitoring dates with the one-step conditional CF (Zhang–Oosterlee 2013); watch the $[a, b]$ width at short maturity.